Fulton–Harris pilot
Detailed formalization plan

Lectures 1–2, items P01–P12, and the exact S_3 example

12 October 2026 / companion to `docs/fulton-harris-plan.tex`

What this document adds. The whole-book plan fixes the pilot's *scope*: 12 items from Lectures 1–2, the S_3 regression example and a choice-free policy. This document turns that scope into work: concrete data structures, proposed Arend signatures, a proof route for each item, the infrastructure lemmas each route needs, a dependency-ordered schedule, and acceptance commands. Every library name below was retrieved from the checkout on 12 October 2026 with the rebuilt CLI (`-ss`, `-fu`, `-ch`). Signatures marked *sketch* are design proposals that have not been typechecked yet.

Summary

- **The M0 blocker is resolved for the pilot cone.** A rebuilt `cli-1.12.0-full.jar` typechecks all 447 modules. A fresh `-no-daemon -r -c` check of the six representation modules exits 0 with no errors (Section 1).
- **The pilot cone is axiom-free by imports.** No representation, linear-algebra, trace, tensor, rational, complex or symmetric-group module can reach `Logic.Classical` or `Logic.Choice`. In the whole library, only `Arith/Complex/FTA/Classical.ard` uses an axiom, and nothing imports it.
- **Replacing `IsIrreducible` is cheap.** Its only consumers are `SchurLemma`, which has no callers, and `PermutationRepresReducible.Not-Irreducible`. For the finite-dimensional pilot, the proposed notion `IsFinIrreducible` tests *split* invariant subspaces, given as an inclusion with a linear retraction, over any `NonZeroRing` (Section 3, D3). It is *not* the general replacement for arbitrary modules. That design is a separate item, D3b, and the name `IsIrreducible` is reserved for it.
- **The work is mostly linear-algebra infrastructure.** The library has bases, coordinates, matrices of maps, change of basis, Smith-form ranks, finiteness of kernels and images, and trace invariance under conjugation. It lacks basis-independent traces, block and Kronecker traces, a basis for Hom, bases of products and tensor products, and trace = rank for idempotents. The plan registers 22 infrastructure items, L01–L22 (L08 split into L08a and L08b) (Section 4).
- **Decidable zero only where it is used.** The definitions are stated over `Ring`, `CRing` or `Field`. `DiscreteField` is a hypothesis only of the kernel and image constructions that test pivots: image splitting (L08b), Schur via kernel and image (P03), and dimension of invariants (P11). Averaging stays over `CRing` with `Monoid.Inv (natCoef card)`.
- **A small constructive-$\mathbb{C}$ bridge is in the pilot.** The explicit S_3 example is transported to `ComplexField` (R8a and R8b). This covers its matrices, decomposition maps, character values and scalar commutant, with explicit inverses for the denominators 2, 3 and 6. Irreducibility over $\mathbb{C}$ gets its own argument, through the scalar commutant and Maschke averaging (L21), which needs no zero test. General scalar-extension machinery stays deferred.
- **One deliberate deviation from the book.** P07 (uniqueness of multiplicities) goes through characters, so it comes after P12. It assumes characteristic 0 through injectivity of `natCoef`, which holds over $\mathbb{Q}$, and it inherits the dependence of P12 on Exercise 1.2. Both are recorded: the relation is `restricted` and the status is `conditional-on-exercises`.

1 Baseline established on 12 October

Check	Result	Status vs. 8 Oct
Rebuild	`./gradlew :cli:jarDep :arend-lib:meta:classes` succeeds; `cli/build/libs/cli-1.12.0-full.jar`; OpenJDK 21.0.10.	Was: unsuitable jar
Whole library	Daemon bootstrap typechecked and serialized all 447 modules in 44.7 s with no `[ERROR]` (loads existing `.arc` where valid).	New
Fresh pilot cone	`arend -no-daemon -r -c` on `Irreducible`, `MaschkeLemma`, `Characters`, `PermutationRepresentation`, `Product`, `LinRepresCat`: exit 0, zero errors, 10.8 s.	Was: parse failure at `Equiv:15:24`
Axioms present	`\axiom` s: `alem`, `achoice` (`Logic/Classical.ard:9,11`); `adchoice`, `acchoice` (`Logic/Choice.ard:6,26`). `alem` has no usages.	Confirmed
Axiom reachability	Import closure (over-approximation) of every pilot module avoids both axiom modules. Only importer of `Logic.Choice`: `Arith/Complex/FTA/Classical.ard:18`, itself unimported.	New; sound because every answer is "no"
Goal reporting	Named goals parse: `{?name}` and `{?name(e)}` (`Arend.g4:262`). Printed as `[GOAL] ... Goal name`. Goals do *not* set a failing exit code; `-c` treats GOAL like ERROR.	New
Daemon caveat	A daemon fixes `-c`, `-r`, `-serialize`, `-L` at start-up and ignores per-request values. Acceptance runs must pass `-no-daemon`.	New
Tests	`test/` has 34 meta/solver tests and no representation tests. Negative tests use the `fails` meta.	Confirmed

Remaining M0 items. (1) Run `arend -no-daemon -r -c` and `-t -r -c` over the whole library, and record the hashes of the jar, `arend-lib/ext` classes and the source tree. (2) Add the two control tests of Section 8. (3) Check in the import-closure audit script under `tools/autoformalization/`. (4) Note one stale comment: `Arith/Complex/FTA.ard:42` names a nonexistent `Arith.Complex.FTA.Choice`.

2 Pilot scope and exit criteria

In scope.

- P01–P12 from the whole-book plan.
- One extra item, P01b: an Arend proof that the old irreducibility notion implies excluded middle.
- The S_3 regression suite: R1–R7 over $\mathbb{Q}$, and R8a/R8b, the explicit transport of the example to constructive $\mathbb{C}$ (structure, then characters).
- The infrastructure L01–L22 that these need.

Out of scope; recorded as deferred.

- General scalar-extension machinery: transfer of arbitrary theorems from $\mathbb{Q}$ to $\mathbb{C}$ (an M2 bridge).
- The general irreducibility notion for arbitrary, possibly infinite-dimensional, modules (D3b).
- The exterior-square conjunct of Prop. 2.1.
- A general decomposition algorithm for Cor. 1.6.
- Completeness of the S_3 character table: that the number of irreducibles equals the number of classes.

- Hermitian forms.
- Proofs of Exercises 1.2 and 2.5, which stay as intentional holes.

Exit criteria. There are two tiers, because the tensor basis (L14) is allowed to finish after everything else.

1. **Core pilot complete.**
   (a) Every P-, L- and R-item has a manifest record whose `statement_review` has a recorded outcome.
   (b) A fresh `-no-daemon -r -c` run of the touched modules reports exactly two goals, `FH_L01_Ex1_2` and `FH_L02_Ex2_5`, and no unregistered goals or errors.
   (c) L14 and the tensor conjunct of P09 may be absent from the source, but their manifest records must say `pending` with the reason. No goal may stand in for them.
   (d) Every declaration outside the dependency closure of the two exercise goals is reported `checked`. Every declaration inside it is reported `conditional-on-exercises`.
   (e) The axiom audit reports no path to `Logic.Classical` or `Logic.Choice`.
   (f) The effort measurements of Section 10 are recorded.
2. **All selected items complete.** The core criteria hold, and in addition L14 and the tensor conjunct of P09 are checked, so no item is `pending`.

Only the second tier is reported as "pilot complete" without qualification.

3 Design decisions

D1. Scalars

- Group-theoretic identities (actions, traces of products, averaging) are stated over `{R : CRing}`. Averaging takes `cardInv : Monoid.Inv (R.natCoef G.card)`, matching the existing `MaschkeLemma` class.
- Definitions use the weakest class that makes them meaningful. The split subrepresentation works over any `Ring`, and `IsFinIrreducible` over any `NonZeroRing`, which excludes the zero ring. The Hom and dual representations need a `CRing`.
- `DiscreteField` is a hypothesis only of constructions that test pivots, that is, kernel and image bases built from `image-fin`, `kernel-fin` and `smith-bases` (`Algebra/Linear/VectorSpace.ard:125--187`). These need a `SmithDomain`; `-ch` confirms that `DiscreteField` is one. See L08 for which of these split over a general `SmithDomain`.
- Results whose proofs avoid pivots are stated over `Field` or `LocalRing`, so that they apply to `ComplexField`. Examples are L21 and the R8a/R8b bridge.
- Concrete examples use `RatField : DiscreteOrderedField Rat` (`Arith/Rat.ard:145`).
- For $|G|^{-1}$ over $\mathbb{Q}$, build the inverse from `RatField.nonZero-Inv` and `natRat/=0` (`Arith/Rat.ard:337`). This is infrastructure lemma L16.

D2. Finite-dimensional representations: truncated versus chosen bases

Two layers are added. The first is the interface; the second does the computing.

```
-- sketch; Algebra/Group/Representation/FinRepres.ard
\class FinLinRepres \extends LinRepres, FinModule     -- isFinModule : ∃ basis (truncated)

\class BasedLinRepres \extends LinRepres {            -- chosen basis, for computation
  | basis : Array E
  | isBasis : IsBasis basis
}
\func BasedLinRepres.toFin (B : BasedLinRepres) : FinLinRepres  -- forget the choice
```

Rules:

- Scalars and dimensions over a `FinLinRepres` are defined only through basis-independence. This follows the `dimension-pair` pattern: a `\level \Sigma` over the truncated basis (`Algebra/Module/FinModule.ard:47`).
- Constructions such as matrices, splittings and averaged projectors take a `BasedLinRepres`.
- No global choice of a basis is made anywhere.

D3. Finite-dimensional irreducibility via split invariant subspaces (pilot notion)

A split subrepresentation is data: a representation `sub`, an intertwiner `incl` into E, and a *linear* retraction of `incl`. The retraction makes `incl` a split monomorphism (`SplitMono`, `Category.ard:61`) in the category of modules `LModuleCat`, not in the category of representations `RepresentationCat`.

```
-- Algebra/Group/Representation/SplitSubLRepres.ard
\record SplitSubLRepres.{u} {R : Ring.{u}} {G : Group.{u}} (E : LinRepres.{u} R G)
  | sub : LinRepres.{u} R G
  | incl : IntertwiningMap sub E
  | split : SplitMono {LModuleCat.{u} R} {sub} {E} incl

-- an invariant split submodule (R^k, incl, retr) gives sub = R^k with g . x := retr (g ** incl x)

\func SplitSubLRepres.fromInvariant (S : SplitSubLModule E P) (P-inv : ...) : SplitSubLRepres E

-- Algebra/Group/Representation/Irreducible.ard (pilot notion; no decidable equality)
```

```
\func IsFinIrreducible.{u} {R : NonZeroRing.{u}} {G : Group.{u}} (E : LinRepres.{u} R G) : \Prop
  => \Sigma (∃ (l : Array E) (0 < l.len) (E.IsBasis l))              -- finite free, positive
    rank
           (\Pi (S : SplitSubLRepres.{u} E) -> (\Pi (x : S.sub) -> x = 0) || IsSurj S.incl)
```

- **Linear, not equivariant, retraction.** A split monomorphism in `RepresentationCat` is an invariant direct summand, and testing only those defines indecomposability. Over $\mathbb{F}_2$, the swap action of C_2 on $\mathbb{F}_2^2$ is indecomposable but not irreducible: the line spanned by $(1,1)$ is invariant and linearly split, but has no invariant complement. The equivariant split is what averaging produces (`maschke-split`, P04a), when $|G|$ is invertible.
- **Universe levels.** `LModuleCat` and `RepresentationCat` take objects at a finite level `u`, so `SplitSubLRepres`, `IsFinIrreducible` and the declarations that quantify over them carry a level parameter `.\{u\}`.
- **No decidable equality in the definition.** The first component is the truncated existence of a basis of positive length; no basis is chosen, and no `dimension` or decision procedure is needed. (An earlier version phrased it as a surjective split inclusion $R^k \to E$. That is equivalent, since such an inclusion is an isomorphism and its invariance condition is automatic, but it misleadingly looked like a statement about subrepresentations.)
- **Nontrivial scalars are required.** Positive rank alone does not exclude the zero module. Over the zero ring, R^1 is already zero, and with `{R : Ring}` the predicate would call it irreducible. Hence `{R : NonZeroRing}`. This is a nontriviality assumption, the negated equality $0 \neq 1$; it is not positive evidence in the apartness or invertibility sense. It needs no decidable equality. The alternative of an explicit nontriviality field on E is listed for review. `DiscreteField` only enters through the constructions that produce split subspaces (L08b, P03).
- **Why this escapes the W_P argument.** A split subspace comes with a retraction. For $W_P \subseteq \mathbb{Q}$ that would mean a linear $r : \mathbb{Q} \to W_P$ with $r \circ j = \mathrm{id}$, so $j \circ r$ is multiplication by an idempotent $c \in \mathbb{Q}$, and $c = 0$ or $c = 1$ decidably. Building that data already settles P.
- **Invariance is a proposition.** It is an equation in a set, so no data has to be chosen to state it.
- **Recorded restrictions.**
  - The notion only makes sense for finite free representations, and it tests only split invariant subspaces of finite rank.
  - It is the pilot's working definition, not the general replacement. Arbitrary and infinite-dimensional modules (for example Verma modules) need the separate design item D3b, which is out of scope here.
  - The name `IsIrreducible` stays unassigned until D3b is decided.
- **Open questions for M1 review.**
  - Should a nonzero-vector witness replace the positive-rank component?
  - `sub` is an arbitrary representation, so finite projective summands are allowed. Rank-one irreducibility then needs a ring without nontrivial idempotents: over $R \times R$, the summand $R \times 0$ is a split subrepresentation. Is this the right generality for D3b?
- **The old definition.** It is renamed `IsStronglyIrreducible` and kept together with its Schur proof, so that P01b can state the excluded-middle consequence. The doc comment says the notion is classical-strength and must not be used as a default.

D4. Representations built from others

```
-- sketch; Algebra/Group/Representation/Hom.ard
\func HomLRepres {R : CRing} {G : Group} (A B : LinRepres R G) : LinRepres R G
  -- carrier LinearMapLModule A B;  (g . phi) v = g ** phi (inverse g ** v)
\func DualLRepres {R : CRing} {G : Group} (A : LinRepres R G) : LinRepres R G
  => HomLRepres A (TrivialAction (RingLModule R) G)
```

```
-- Algebra/Group/Representation/Tensor.ard
\func TensorLRepres {R : CRing} {G : Group} (A B : LinRepres R G) : LinRepres R G
  -- carrier TensorProductModule A B; g acts by tensorLift of (a, b) |-> g a ⊗ g b
```

- The direct sum is the existing `ProductLRepres`.
- Finite indexed sums are iterated `ProductLRepres`. They are only needed in the S_3 example and in P07, both with a small fixed number of summands.
- The library's misspelled `InterwiningMap` (and the module of that name) was renamed to `IntertwiningMap`, overriding the whole-book plan's instruction to keep the existing spelling.

D5. Characters

- The existing `Character E bv g` becomes the *based* character.
- The basis-free `character` on a `FinLinRepres` is defined by eliminating the truncated basis into a level-(-1) Σ-type, as `dimension-pair` does. This needs L05.
- Class functions are `G -> R` with `\Pi g h -> f (h * g * inverse h) = f g`.
- The pairing is the inverse-character one: $\langle \chi, \psi \rangle = |G|^{-1} \sum_g \chi(g^{-1})\psi(g)$.
- Comparing with the book's conjugate pairing belongs to the deferred $\mathbb{C}$ bridge.

D6. Exercise holes

Exercises 1.2 and 2.5 are stated in full. Their bodies are the named goals `{?FH_L01_Ex1_2}` and `{?FH_L02_Ex2_5}`, and each declaration's doc comment cites the exercise. Consumers are written against the statement and inherit `conditional-on-exercises`.

- **To be checked in M0.** The checker must still elaborate a consumer whose dependency has a goal body.
- **Fallback.** If it cannot, the consumer takes the exercise statement as an explicit hypothesis. Its name gets the suffix `-cond`; for example, the conditional form of `orthogonality` is `orthogonality-cond`.

D7. Placement and names

New declarations go in mathematical modules under `Algebra/Group/Representation/` and `Algebra/Module/`, and they follow the naming table of the `arend-formalize` skill. Book numbers appear only in doc comments (`- | Fulton-Harris Prop.~1.5`) and in the manifest. The S_3 objects that are reusable (the sign representation and the standard representation of S_n for $n = 3$) go in the library. The S_3 numerical checks go in `test/Algebra/Representation/S3Test.ard`.

4 Infrastructure backlog

Each L-item is a lemma or construction missing from the checkout. Its absence was confirmed by symbol search, usages and signature search. Sizes are rough proof-effort classes: S is under 30 lines, M is 30–150, L is over 150.

ID	Statement (sketch)	Reuse	Size	Consumers
L01	`toLinearMap_*`: `E.toLinearMap (g * h) = E.toLinearMap g ∘ E.toLinearMap h`	`**-assoc`, `exts`	S	L05, P08, P11
L02	Matrices of the action reverse products: `toMatrix lv bv (toLinearMap (g*h)) = toMatrix .. h product toMatrix .. g`. So $g \mapsto$ matrix of $\rho(g)$ is an *anti*homomorphism into `MatrixRing`; the homomorphism is $g \mapsto$ matrix of $\rho(g^{-1})$, or a map from the opposite group	L01, `toMatrix_*` (`LinearMap.ard:109`; row convention)	S	P01, P08, P09
L03	`toMatrix` is linear in the map: `toMatrix_+`, `toMatrix_*c`, `toMatrix_BigSum`, `toMatrix_FinSum`	`basis-split-unique`, `linearMap_BigSum`	M	P11
L04	Trace is basis independent: `Trace (toMatrix lu bu f) = Trace (toMatrix lu' bu' f)` for an endomorphism	`change-basis_matrix-left`, `-right`, `toMatrixInv`, `basis-iso`, `Trace-conjugation` or `Trace-prod`	M	L05
L05	`endoTrace {R : NonZeroCRing} (U : FinModule R) (f : LinearMap U U) : R`, defined by the `dimension-pair` pattern, with `endoTrace-char`	L04	S	P08, P11
L06	Trace of a block-diagonal matrix: `Trace (blockMatrix A B) = Trace A + Trace B`	`blockMatrix`, `elem00/11` (`Matrix.ard:51`)	S	P09
L07	Basis of `ProductLModule`: `in1`-images followed by `in2`-images; matrix of a product map is `blockMatrix`	`IsBasis`, `basis-split-unique`	M	P09
L08a	**Kernel splits over a `SmithDomain`.** With `smith-bases`, $f(u'_j) = d_j v'_j$, and $\ker f$ is spanned by the u'_j with $d_j = 0$, because a `SmithDomain` is a domain and $d_j = 0$ is decidable. Projection onto those coordinates is a retraction.	`smith-bases`, `kernel-fin`	M	P03, P04b
L08b	**Image splits only with invertible pivots.** Over a general `SmithDomain` it fails: $2\cdot : \mathbb{Z} \to \mathbb{Z}$ has image $2\mathbb{Z}$, which has a basis but no retraction from $\mathbb{Z}$. So the image of f is split if every nonzero Smith coefficient d_j is invertible; this holds automatically over a `DiscreteField`. The retraction sends $v'_j \mapsto d_j^{-1} e_j$ for $j <$ rank.	`smith-bases`, `image-fin`	M	P03

ID	Statement (sketch)	Reuse	Size	Consumers
L09	Over a `DiscreteField`: `retr ∘ incl = id` and `dim = T.dim` (a full split subspace) imply `IsSurj incl`	`toMatrix_*`, `matirx-inv` (sic, `Matrix.ard:542`), `matrix-split_<=`	S	P03
L10	Dimension bound: a split inclusion has `dim <= T.dim` over a `NonZeroCRing`	`matrix-split_<=` (`Matrix.ard:645`)	S	L21, R3
L11	Basis of `LinearMapLModule U V` for based U, V: matrix units; upgrade `matrix-equiv` (set-level, `LinearMap.ard:134`) to a `ModuleIso`	L03	M	P02, P12
L12	Kronecker product of matrices and `Trace (kron A B) = Trace A * Trace B`	`BigSum` double-sum lemmas	M	P09, P12
L13	Matrix of $\varphi \mapsto b \circ \varphi \circ a$ in the matrix-unit basis is `kron (transpose A) B`	L11, L12, `Trace-transpose`	M	P12
L14	Basis of `TensorProductModule` of based modules $(e_i \otimes f_j)$; tensor of linear maps; matrix is a Kronecker product	`tensorLift`, `tensorLift-unique`	L	P09
L15	Trace of an idempotent: for $p \circ p = p$ on a based module over `DiscreteField`, `endoTrace p = natCoef (dimension (image-fin p))`	L08, L07, `image-fin.dimension_rank`	M	P11
L16	`Inv (RatField.natCoef (suc n))`; injectivity of `natCoef` over `RatField`	`nonZero-Inv`, `natRat/=0`	S	P04, P07, R*
L17	`SymFinGroup (n : Nat) : FinGroup (Sym n)` from `SymmetricGroup` and `SymFin` (no instance exists today); check that the `DecSet` fields agree and that `EquivFin` does not win instance search	`Algebra/Group/Symmetric.ard: 65,284,297`	S	R1–R7
L18	Natural action `SymAction n : GSet (SymmetricGroup n)` on `Fin n`. Products are left-to-right (`(e * e') j = e' (e j)`, `Equiv.ard:233`), so the action must be `e ** i = inverse e i`; then `PermRepr` gives $(g \cdot f)(j) = f(g\,j)$	`symQEquiv`, `Sym.equals`	S	R2, R4, R5
L19	Fixed points as a decidable subset of a finite `GSet` with decidable equality, with a `FinSet` instance (`SigmaFin.DecSubSet-isFin`); conjugacy classes of an explicit finite group as decidable subsets	`DecSubSet`, `SigmaFin`, `FinSum-inj`	S	P10, R1
L20	Inversion $g \mapsto g^{-1}$ as a `QEquiv` for every `Group`, and the reindexing `FinSum (x o inverse) = FinSum x` (today only `SymmetricGroup.inv-isEquiv` exists)	`inverse-isInv`, `FinSum_Equiv`	S	P09, P11, P12

ID	Statement (sketch)	Reuse	Size	Consumers
L21	**Irreducibility from a scalar commutant.** Over a commutative `LocalRing` R (hence `NonZeroRing`), with `cardInv`: if $E \cong R^n$ with $n > 0$ (a full split subspace T with `0 < T.dim`) and every intertwiner $E \to E$ is scalar, then `IsFinIrreducible E`. Positive rank is essential, since the zero representation also has only scalar endomorphisms. Route: a split invariant S gives the equivariant idempotent $\pi = \mathit{incl} \circ \rho$ (P04a). By the commutant $\pi = c \cdot \mathrm{id}_E$, so $(c^2 - c)\,\mathrm{id}_E = 0$. Evaluating at $T.\mathit{incl}(e_0)$ and applying $T.\mathit{retr}$ gives $(c^2 - c)e_0 = 0$ in R^n, hence $c^2 = c$; this step uses $n > 0$. Then `LocalRing.connected` (`Algebra/Ring/Local.ard:37`) gives $c = 0$ (so `dim = 0`, using $0 \neq 1$) or $c = 1$ (so `incl` is surjective). No zero test is used, so it applies to `ComplexField`.	P04a, `LocalRing.connected`, L10	M	R3, R8a
L22	Ring homomorphism $\mathbb{Q} \to \mathbb{C}$: `fromRealRingHom` (`Arith/Complex.ard:108`) composed with the structure map of `RealField : OrderedFieldAlgebra RatField Real`; transport of matrices (`matrix-map`), of traces, and of `Monoid.Inv` along a ring homomorphism, which gives the inverses of $2, 3, 6$ in `ComplexField`	`fromRealRingHom`, `RingHom`, `charPoly_map` pattern	S	R8a, R8b

Notes.

- L14 is the largest item. The pilot needs the tensor character formula, so L14 is scheduled early, in wave 1, as an independent track. If it stalls, only the tensor conjunct of P09 is held up; P12 uses the Hom route (L11–L13) instead.
- For L14, `Algebra/Algebra/TensorProduct.ard` builds a bilinear multiplication by nesting `tensorLift`. That is the template for the tensor $f \otimes g$ of two linear maps. Note that `tensorLift_tensor` is propositional, not `idp`.
- L03 can reuse `AddMonoidHom.func-FinSum` (`Algebra/Monoid/MonoidHom.ard:92`), which applies to linear maps.
- L08 is the design-critical item, and it is split in two because the image half is false over a general `SmithDomain` (the counterexample $2\mathbb{Z} \subseteq \mathbb{Z}$ is in L08b). It is prototyped and typechecked first, together with D3 and P03.
- The image of an *idempotent* is always split, with π itself as the retraction onto its image. So P04, P11 and L21 never need L08b.

5 Item specifications

Each item lists the source statement, the constructive target, the relation to the source (`same`, `reformulated`, `restricted`), the proposed declarations, the proof route, dependencies and acceptance. Page numbers are printed pages; the physical PDF page is the printed page plus 15, an offset checked on pp. 3–16.

P01 Representations, based finite representations, split subobjects, irreducibility

Source. §1.1, pp. 3–4. A representation is a homomorphism $\rho : G \to \mathrm{GL}(V)$. V is irreducible if it has no proper nonzero invariant subspace.

Target.

- D2 (`FinLinRepres` and `BasedLinRepres`).
- D3 (`SplitSubLRepres` and the pilot notion `IsFinIrreducible`).
- The bridge to $G \to \mathrm{GL}(V)$, with the order fixed. `LinearMapRing` multiplies as `f * g = g âĹŸ f` (`Algebra/Module/Instances.ard:28`), while an action satisfies $\rho(gh) = \rho(g) \circ \rho(h)$ (L01). So $g \mapsto \rho(g)$ is an *anti*homomorphism into the units of `LinearMapRing E`. The bridge is therefore `LinRepres.toGroupHom : GroupHom G (units of LinearMapRing E)` with $g \mapsto \rho(g^{-1})$. The equivalent alternative, a homomorphism from the opposite group with $g \mapsto \rho(g)$, is recorded for review. Either way, the construction comes with its inverse.

Relation: `reformulated`. Invariant subspaces are replaced by split ones.

Route.

- The induced action on R^{dim} is $g \cdot x = retr(g \cdot incl\, x)$.
- Its `**-assoc` uses `incl-invariant` and `retr_incl`.
- The bridge reuses `toLinearMap`, `toLinearMap_ide`, L01 and `inverse_*`. A regression test pins the direction of the homomorphism.

Migration.

- Restate `PermutationRepresReducible.Not-Irreducible` (`PermutationRepresentation.ard:57--67`) for the new notion, over an arbitrary ring and an arbitrary `GSet X`, with the same hypotheses as now. The old theorem's conclusion `0 = {R} 1` turns into the hypothesis that R is a `NonZeroRing`, and the new conclusion is `Not (IsFinIrreducible (PermRepr X))`.
- The constant line $j(c) = (c, c, \dots)$ with $r(f) = f(x_0)$ is a split invariant subspace of dimension 1 for any X. `nonConstantVector` shows that j is not surjective. The refutation only uses the second component of `IsFinIrreducible`.
- **Recorded change of strength.** For infinite X, the new predicate's first component, finite freeness, generally fails anyway. The migrated theorem is then weaker than the old one, which refuted irreducibility in the strong sense for every X. The general-module form waits for D3b.
- `SubLRepres.IsTrivial` stays, documented as the strong dichotomy.

Acceptance. One-dimensional examples are irreducible over any `NonZeroCRing`: the trivial and sign representations (L10 gives $dim \leq 1$). The migrated `Not-Irreducible` checks.

P01b The old notion implies excluded middle

Target. `IsStronglyIrreducible.excludedMiddle`: from `IsStronglyIrreducible (TrivialAction (RingLModule R) G)` and $0 \neq 1$ in R, derive `\Pi (P : \Prop ) -> P || Not P`.

Route.

- The submodule is W_P with `contains x => P || x = 0`. Its closure fields are short case splits.
- Then follow the two cases of the whole-book plan, page 4.
- This is an ordinary constructive theorem. It introduces no axiom and documents why D3 is needed.

Size. S. It is a good first task for calibrating the workflow.

P02 Direct sum, dual and Hom representations; Exercise 1.2 (hole)

Source. §1.1, pp. 4–5; Exercise 1.2.

Target. `HomLRepres` and `DualLRepres` (D4); the existing `ProductLRepres`. Exercise 1.2 as an intentional hole:

```
-- | Fulton--Harris Exercise 1.2: invariant vectors of Hom(V, W) are the G-linear maps.
\lemma invariantHom {R : CRing} {G : Group} {A B : LinRepres R G} (f : LinearMap A B)
  : (\Pi (g : G) -> g HomLRepres.** f = f) <-> (\Pi (g : G) (a : A) -> f (g ** a) = g ** f a)
  => {?FH_L01_Ex1_2}
```

Relation: `same`.

Route. The action fields of `HomLRepres` are pointwise, using `exts`, `**-assoc` and `inverse_*`. The pattern to follow is `IntertwiningMapLModule` in `LinRepresCat.ard`.

Acceptance. `HomLRepres` and `DualLRepres` check, and only the registered goal remains.

P03 Schur's lemma (1)

Source. Schur 1.7(1), p. 7.

Target.

```
\lemma SchurLemma {K : DiscreteField} {G : Group} {A B : BasedLinRepres K G}
  (p : IsFinIrreducible A) (q : IsFinIrreducible B) (f : IntertwiningMap A B)
  : f.IsZero || IsEquiv f
```

Relation: `reformulated`. It uses the pilot irreducibility. As implemented in wave 1 it concludes `IsEquiv f`, like the old lemma, rather than `ModuleIso`; `ModuleIso.iso<->inj+surj` converts between the two. The `DiscreteField` hypothesis is needed for the image splitting (L08b), not for the definitions.

Route.

- The kernel of f is split by L08a, which works over any `SmithDomain`. The image is split by L08b, which needs invertible pivots, so this is where `DiscreteField` is used. The target is a proposition, so the truncation in `smith-bases` can be eliminated.
- Invariance comes from `KernelLRepres` and `ImageLRepres`.
- Then copy the case structure of the existing `SchurLemma` (`Irreducible.ard:14--21`), using L09 for "dim equal implies surjective".

Risk. This item validates D3, so it is prototyped first, typechecked together with L08a and L08b. If the image splitting cannot be stated cleanly, revise D3 before writing anything else.

Not covered. Schur over constructive $\mathbb{C}$. Specific complex maps can have explicitly known kernel and image bases. What is missing is a general procedure that constructs such bases, and their retractions, without additional evidence such as invertible pivots. So the $\mathbb{C}$ side of the pilot uses L21 instead.

P04 Maschke: invariant complements

Source. Prop. 1.5, p. 6.

Target. Two declarations.

- `P04a` over `CRing`, with $|G|^{-1}$. Given a split invariant subspace S, it returns an equivariant idempotent $\pi = \mathit{incl} \circ \rho$, where ρ is the averaged retraction, with $\pi \circ \mathit{incl} = \mathit{incl}$. Its image is S, and it needs no basis and no zero test, so it also applies to `ComplexField`. L21 uses it.
- `P04b` over `DiscreteField`. It returns a complement S' as a `SplitSubLRepres` together with an isomorphism of representations $E \cong S \oplus S'$.

Relation: `restricted`. The input is split rather than an arbitrary subspace; the whole-book plan, page 6, explains why.

Route.

- Average $p_0 = \mathit{incl} \circ \mathit{retr}$. Reuse `SumOverGroup`, `FinSumRewrite` and the `mean-func-preserve` argument (`MaschkeLemma.ard:28--45`), generalized from a `SubLRepres` target to `E`.
- The complement is $\ker \pi$, split by $\mathrm{id} - \pi$. L08a supplies its basis over any `SmithDomain`; `DiscreteField` is stated for uniformity with P03, and weakening it is a review experiment.
- The isomorphism is `coprod-map` of the two inclusions, with inverse $(\pi, \mathrm{id} - \pi)$.

Regression. Characteristic $p \mid |G|$: over `FinField` with $p = 2$ and $G = S_3$, `cardInv` is unavailable. A test documents that the hypothesis is needed.

P05 Complete reducibility for the presented inputs

Source. Cor. 1.6, p. 6.

Target. An explicit decomposition of the permutation representation of S_3 on $\mathbb{Q}^3$ as the trivial line plus the standard representation. It is an isomorphism of representations with both summands proved irreducible. Relation: `restricted`; the general corollary is deferred.

Route. Apply P04b to the split line $(1, 1, 1)$, or write both inclusions out explicitly. Irreducibility of the standard representation is R3.

P06 Schur's lemma (2): scalar commutants

Source. Schur 1.7(2), p. 7.

Target.

- `IsSplitIrreducible E`: `IsFinIrreducible` plus "every `IntertwiningMap E E` is a scalar multiple of the identity".
- A proof that one-dimensional representations are split irreducible.
- A proof that the standard representation of S_3 is split irreducible over any commutative `LocalRing` in which 6 is invertible; L21 needs that inverse anyway. A 2×2 matrix commuting with $\rho(12)$

and $\rho(123)$ is scalar, by solving linear equations. Every division is meant to be by a divisor of 6; confirm this when the equations are written out. The representation has rank $2 > 0$, which L21 requires. Irreducibility then follows from L21. Instantiate at `RatField` (R7) and at `ComplexField` (R8a).

Relation: `restricted`. The book's eigenvalue argument needs $\mathbb{C}$ to be algebraically closed. Here the scalar commutant is proved per example; the general version is the M2 bridge.

P07 Uniqueness of multiplicities

Source. Prop. 1.8, p. 7.

Target. Take pairwise non-isomorphic split irreducibles V_i and two decompositions $V \cong \bigoplus V_i^{a_i} \cong \bigoplus V_i^{b_i}$, given as iterated `ProductLRepres`. Then $a = b$. Relation: `restricted`; it assumes characteristic 0 through injectivity of `natCoef` (L16).

Route (deviation).

- Characters are additive (P09) and the irreducible characters are orthonormal (P12). Together these give $\langle \chi_{V_i}, \chi_V \rangle = a_i = b_i$ in K; L16 then gives equality in $\mathbb{N}$.
- The book's route, through Hom-dimensions, needs additivity of $\dim \mathrm{Hom}_G$ over direct sums. The character route reuses work the pilot does anyway.

Dependencies: P09, P12, L16. Through P12, P07 depends on the Exercise 1.2 hole, so its status is `conditional-on-exercises` until that hole is closed; the characteristic-0 restriction appears in its signature.

P08 Basis-independent character; class function; $\chi(1) = \dim$

Source. §2.1, p. 13.

Target.

- `character (E : FinLinRepres R G) : G -> R`, defined through L05.
- `character-char`: it equals `Character E bv` for every basis.
- `character_conj`: `character E (h * g * inverse h) = character E g`.
- `character_ide`: the value at 1 is `natCoef (dimension E)`.

Relation: `same`.

Route.

- Class invariance: L02 followed by `Trace-prod`.
- The value at 1: `Character.of-1` (existing) and `dimension-unique`.

P09 Characters of sums, tensors and duals

Source. Prop. 2.1, p. 13.

Target.

- $\chi_{V \oplus W} = \chi_V + \chi_W$, using L06 and L07.
- $\chi_{V \otimes W} = \chi_V \chi_W$, using L12 and L14.
- $\chi_{V^*}(g) = \chi_V(g^{-1})$ and $\chi_{\mathrm{Hom}(V,W)}(g) = \chi_V(g^{-1})\chi_W(g)$, using L11, L13 and `Trace-transpose`.

The exterior-square conjunct is a registered pending item, not a hole. Relation: `reformulated` for the dual. The book writes $\overline{\chi_V}$; over $\mathbb{C}$ this agrees with $\chi_V(g^{-1})$ only after the roots-of-unity argument, which is deferred.

P10 Permutation characters count fixed points; Exercise 2.5 (hole)

Target.

```
-- | Fulton--Harris Exercise 2.5 (the original fixed-point formula).
\lemma permCharacter {R : CRing} {G : Group} (X : FinGSet G) (g : G)
  : Character (PermRepr X) (deltaBasis X) g
      = R.natCoef (FinSet.card (FixedPointsFin X g))
  => {?FH_L02_Ex2_5}
```

- The supporting definitions are `deltaBasis` (the indicator functions) and `FixedPointsFin` (the decidable subset).
- Both definitions are proved. Only the formula itself is the hole.

Relation: `same`. Consumers are R5 and the character of the regular representation (M2).

P11 Averaging onto invariants; $\dim V^G$ as a trace

Source. Prop. 2.8 and (2.9), pp. 15–16.

Target.

- `averageMap` $\varphi = |G|^{-1}\sum_g \rho(g)$, over `CRing`.
- φ is equivariant and idempotent, and its image is exactly the set of invariant vectors.
- Over `DiscreteField`: `natCoef (dimension (invariants E))` equals $|G|^{-1}\sum_g \chi_E(g)$.

Relation: `same` for Prop. 2.8. For (2.9) it is `reformulated`: the equation holds in K, and equality of natural numbers needs L16.

Route.

- Use the `FinSum` of `toLinearMap g` in `LinearMapAbGroup`, reindexed with `FinSum_Equiv` and `Group.translate-is-Equiv`.
- Trace commutes with the sum by L03 and the linearity of `Trace`.
- Dimension equals trace by L15.

P12 Orthogonality of split irreducible characters

Source. Thm. 2.12 and (2.10), p. 16.

Target. Two declarations, for split irreducibles V, W:

- `orthogonality-iso`: from `ModuleIso` intertwining $V \cong W$, conclude $\langle \chi_V, \chi_W \rangle = 1$.
- `orthogonality-noniso`: from `Not` (such an isomorphism), conclude $\langle \chi_V, \chi_W \rangle = 0$.

Relation: `restricted`. The input is split irreducibles, and the conclusion splits into two cases rather than a decided dichotomy.

Route.

- Apply P11 to `HomLRepres V W`. The invariants are the intertwiners by Exercise 1.2 (hole), so this item is `conditional-on-exercises`.

- In the non-isomorphic case, P03 forces every intertwiner to be zero, so the dimension is 0.
- In the isomorphic case, P06 gives the intertwiners a basis $\{\mathrm{id}\}$, transported along the isomorphism, so the dimension is 1.
- The character of `HomLRepres` comes from P09.

6 The exact S_3 regression suite

All objects are stated over an abstract commutative ring or `LocalRing` and instantiated at `RatField` and at `ComplexField`. The group is G = `SymFinGroup 3` (L17). The generators are $\sigma = (1\,2)$ (`transposition`) and $\tau = (1\,2\,3)$ (`cycle`, `Symmetric.ard:139`). The sign is `signHom` (`Symmetric.ard:574`).

ID	Statement	Route
R1	The six elements, listed explicitly, are pairwise distinct and exhaust S_3. The class sizes are $1, 3, 2$; each class is a decidable subset with its `card` computed. `card = fac 3` reduces to 6.	L17, L19; decidable equality from `SymFin`; `Sym.equals`
R2	Representations: trivial; sign ($g \cdot x = \mathrm{sgn}(g)x$, a new `CharacterLRepres` for a `MonoidHom` into the multiplicative monoid, using `signHom` and `sign_*`); `PermRepr` on `Fin 3`; standard ($e_1 - e_2$, $e_2 - e_3$, as a `SplitSubLRepres` of `PermRepr`). The standard representation is given by an explicit inclusion $R^2 \to R^3$ and an explicit retraction, so it needs only the D3 definitions (wave 1), not the P01 bridge or the product basis L07.	D3, L18
R3	The standard representation is irreducible, by L21 applied to the scalar commutant (R7). A split invariant subspace gives an equivariant idempotent (P04a), which is a scalar c with $c^2 = c$, so $c = 0$ or $c = 1$ by locality. The same proof works over `ComplexField` (R8a). *Rejected route:* excluding a rational eigenvector of τ needs a proof that $\lambda^2 + \lambda + 1$ has no rational root, for example by ordered-field positivity; `equation.cRing` alone cannot do it, and the route does not transfer to $\mathbb{C}$.	R7, L21, P04a
R4	`PermRepr` is isomorphic to trivial $\oplus$ standard (P05).	P04b or explicit maps
R5	Character rows: trivial $(1, 1, 1)$, sign $(1, -1, 1)$, standard $(2, 0, -1)$ on $[1], [\sigma], [\tau]$; permutation $(3, 1, 0)$, also as P10's consequence (conditional).	Explicit 2×2 traces
R6	Orthogonality table: the nine pairings $\langle \chi_i, \chi_j \rangle$ computed by exact arithmetic with weights $1, 3, 2$ and $\lvert G \rvert^{-1} = 1/6$.	L16, normalization in `Rat`
R7	Scalar commutant of the standard representation over a commutative `LocalRing` with 6 invertible (P06), instantiated at `RatField`; and the Hom-dimension consequences used in P12. Proved before R3 and R8a, which depend on it.	Linear equations over an abstract ring
R8a	**Constructive-$\mathbb{C}$ bridge, structural stage.** Transport R2's matrices along the ring homomorphism $\mathbb{Q} \to \mathbb{C}$ (L22). This gives the decomposition maps of R4, with their inverses, and $\lvert G \rvert^{-1}$ and the other denominators as images of rational inverses, so no zero test on $\mathbb{C}$ is used. The scalar commutant of R7 is instantiated at `ComplexField`. Irreducibility over $\mathbb{C}$ gets its own proof through L21; it is *not* inferred from irreducibility over $\mathbb{Q}$, since irreducibility does not transfer in general.	L21, L22, P04a, R4, R7
R8b	**Constructive-$\mathbb{C}$ bridge, character stage.** The character values of R5 over `ComplexField`, as images of rationals under L22 (traces commute with ring homomorphisms), and the R6 pairings over $\mathbb{C}$.	R5, R6, R8a, L22

Arithmetic risk; abstract-then-instantiate is mandatory.

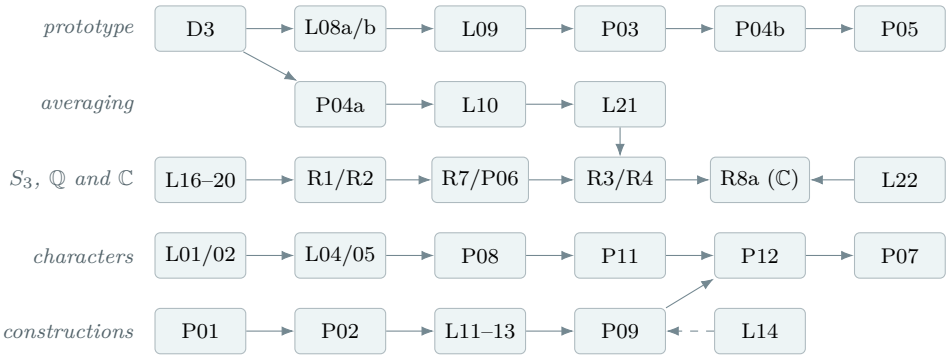
- `Rat` values are normalized by gcd, so closed `Rat` identities may not reduce to `idp`.
- Concrete `Complex` arithmetic unfolds into `re`/`im` components. `equation.cRing` then fails even on `x * y = y * x`, and `rewrite` cannot match under `*` (failure mode 16 of the `arend-prove` skill).
- Therefore R2–R8b are stated and proved over an abstract `{R : CRing}` or `LocalRing`, with explicit inverses of the needed integers as parameters. They are then instantiated twice, at `RatField` and at `ComplexField`.

- Concrete-field proofs are limited to supplying those inverses (L16, L22) and to the instantiation step.

7 Dependency order and schedule

The work runs in waves. Items inside a wave are *not* all independent: each wave lists its items as chains, written $a \to b$, and $a \mid b$ separates items that can run in parallel. An item may only start once everything before it in its chain, and every earlier wave it depends on, has checked.

The recommended start is a typechecked prototype of D3, L08 and Schur (wave 1). Wave 2 then tests the interfaces on the S_3 decomposition and scalar commutant, over both $\mathbb{Q}$ and constructive $\mathbb{C}$, before the general character theory is built.



The diagram shows the main chains; the item specifications list every dependency. P12 also depends on P03 and on R7/P06, and R8b (the $\mathbb{C}$ character stage) follows R5 and R6; these edges are not drawn. The dashed arrow marks L14, which feeds only the tensor conjunct of P09 and runs as a separate track.

Wave	Items	Gate to the next wave
0	Finish M0 (Section 1) | control tests | manifest skeleton for all P/L/R items | statement review of D3, L08a, L08b, P03	D3 statement review signed off
1	**Prototype:** D3 definitions (`SplitSubLRepres`, `IsFinIrreducible` over `NonZeroRing`, old notion renamed) → (L08a | L08b | L09) → P03; everything typechecked with no goals	Schur typechecks; D3 stands or is revised
2	**Interface test on S_3 over $\mathbb{Q}$ and $\mathbb{C}$.** Chain A: (L16 | L17 | L18 | L19 | L20) → R1 → R2 → P06/R7. Chain B, in parallel with A: P04a → L10 → L21 (generic; no S_3 input). Chain C, after A and B: R3 → R4 (with explicit decomposition maps, since P04b comes in wave 3). Chain D, after C: L22 → R8a	Decomposition, commutant and irreducibility check over both fields, with no zero test on $\mathbb{C}$
3	L01 → ((L02 → L04 → L05 → P08) | (P01 bridge with $\rho(g^{-1})$, migration → P02 with hole)); in parallel: P01b | (P04b → P05) | L03	Characters and Maschke complements check
4	(L06 | L07 | P10 with hole) → R5 → R8b-values; (L11–L13 | L15) → P11	Invariant-dimension formula and $\mathbb{C}$ character values check
5	P09 (sum, dual, Hom conjuncts) → P12 → (P07 | R6 → R8b-pairings)	**Core pilot complete** (Section 2)
Track T	L14 → tensor conjunct of P09, from wave 1 onwards	**All selected items complete**

8 Verification protocol

Edit loop. Fast checks

Run checks from `arend-lib/`, through the daemon: `arend -serialize <Module>[:<def>]`. Look up names with `-ss`, `-ps`, `-fu` and `-ch` before writing any helper. After a class-level change, the daemon has to be refreshed with `arend -daemon-refresh`. The `arend-formalize` and `arend-prove` skills describe this loop.

Acceptance. Fresh checks

```
cd arend-lib
arend --no-daemon -r -c Algebra.Group.Representation.Irreducible \
  Algebra.Group.Representation.SplitSubLRepres Algebra.Group.Representation.Hom \
  Algebra.Group.Representation.Characters Algebra.Group.Representation.MaschkeLemma \
  Algebra.Group.Representation.Invariants ...           # every touched module
arend --no-daemon -t -r -c                             # library tests, incl. S3Test
```

The acceptance script parses the `[GOAL]` and `[ERROR]` lines.

- The run passes only if the goal set is exactly {`FH_L01_Ex1_2`, `FH_L02_Ex2_5`} at the manifest's declarations, and there are no errors.
- Goals alone do not fail the exit code, so the script has to check the goal set itself.
- Under `-c`, the double checker reports goals like errors. Those lines are matched against the manifest too.

Controls. Prove that the harness can fail

Two tests go under `test/Algebra/Representation/`.

1. A deliberately wrong proof of `0 = {Nat} 1`, wrapped in the `fails` meta. The control is that removing `fails` must turn the run red.
2. A small module that imports `Logic.Classical` and uses `alem`. The axiom audit must flag it.

Axiom audit. Two levels

- **Level 1, the gate.** The import-closure script reports any path to `Logic.Classical` or `Logic.Choice`. It over-approximates real dependency, so a "no" is conclusive.
- **Level 2, investigation.** If Level 1 ever says "yes", `-fu alem`, `-fu achoice`, `-fu acchoice` and `-fu adchoice` locate the actual uses.
- **Hypothesis review.** Reviewers also check signatures for classical strength: any hypothesis of the shape `\Pi (P : \Prop ) -> ...` or a decision procedure for an undecidable property.

Semantic review. Per item

- Check that the formal statement matches the source in quantifiers, scalar class, finiteness, and isomorphism versus equality.
- Record the relation and the data that replaces each classical step.
- Run the Phase-4 finish checklist of the `arend-formalize` skill: goals, names, comments, placement, hypothesis weakening, redundancy and layout.
- Try weakening each hypothesis, in particular `DiscreteField` to `Field` or `CRing`, and record which ones the checker accepts. Each successful weakening is a better statement.

9 Manifest records

Location. One JSON Lines file, `arend-lib/docs/fulton-harris/manifest.jsonl`, with one record per P-, L- and R-item. It uses the whole-book plan's schema and states. Two representative records, shown in YAML for readability:

```
id: FH.L01.Schur1.7.1            # P03
kind: theorem
source: {printed_page: 7, pdf_page: 22, label: "Schur's Lemma 1.7 (1)"}
original: "A G-map between irreducibles is zero or an isomorphism."
constructive: "For split-irreducible based representations over a discrete field,
               an intertwiner is zero or a module isomorphism."
source_relation: reformulated
formal_target: Algebra.Group.Representation.Irreducible:SchurLemma
dependencies: [GAP.L08.smithSplit, GAP.L09.splitSurj, KernelLRepres, ImageLRepres]
status: planned
statement_review: pending
axiom_policy: no_excluded_middle_or_choice

id: GAP.L15.idempotentTrace
kind: infrastructure
statement: "endoTrace p = natCoef (dimension (image-fin p)) for p idempotent"
formal_target: Algebra.Linear.VectorSpace:idempotent-trace   # placement to review
consumers: [FH.L02.Prop2.8, FH.L02.Thm2.12]
acceptance: "checked with no goals; no new hypotheses beyond DiscreteField"
status: planned
```

Status after a check. Once an item is checked, its record gains the jar hash, the source revision, the command, the exit status, the goal lines and the reviewer outcome.

10 Measurement, risks and decisions needed

Measurement. What to record per item

- Task class: wrapper, routine algebra, bridge, or major theorem.
- Repair cycles used, out of a budget of five.
- Wall-clock checking time.
- Human review minutes.
- Lines added and lines of reused library code.
- The outcome classification: success, source ambiguity, missing lemma, elaboration failure, or search failure.

After wave 2, re-estimate the remaining waves from the observed medians. The pilot's purpose is calibration, so these records are part of the deliverable.

Risk	Early signal	Response
D3 too weak for Schur	The wave-1 prototype cannot state L08a/L08b cleanly, or P03 needs extra hypotheses	Revise D3 before wave 2; candidates: projective (finitely generated, split) instead of free submodules
Complex bridge stalls	R8a or R8b needs a zero test, or concrete `Complex` terms block unification	Keep every S_3 statement abstract (Section 6); supply only inverses of $2, 3, 6$ through L22; record any remaining step as a named gap, never as a hidden decision procedure
Truncation friction	`character` or `endoTrace` cannot eliminate `isFinModule`	Follow `dimension-pair` exactly; fall back to `BasedLinRepres`-only statements for the pilot
Consumers of holes do not elaborate	Errors at use sites of `invariantHom`	Use the `-cond` hypothesis form (D6)
Rational arithmetic is slow or stuck	R5/R6 time out or `equation` fails on literals	Abstract-field statements, instantiate at the end (Section 6)
L14 (tensor basis) overruns	No progress after one wave	Keep it as `pending`: the pilot can reach "core complete" but not "all selected items complete" (Section 2); P12 is unaffected
Order conventions	Three library conventions compose diagrammatically: `toMatrix_*` (row convention), `LinearMapRing` (`f * g = g ∘ f`) and `SymmetricGroup` (`(e * e') j = e' (e j)`). As a consequence, $g \mapsto \rho(g)$ is an antihomomorphism into `LinearMapRing`	Name lemmas by mathematical order; add a regression test that fixes each convention
Daemon flag locking	A "fresh" run silently reuses caches	Acceptance uses `-no-daemon` only

Decisions. Needed from the project owner before wave 1

1. Approve D3 as the *pilot* notion `IsFinIrreducible`, over any `NonZeroRing`, together with its recorded restrictions and the three open questions listed under it.
2. Confirm that the name `IsIrreducible` stays reserved for the general-module design D3b, and approve renaming the old notion to `IsStronglyIrreducible` instead of deleting it.
3. Choose the bridge direction for P01: $g \mapsto \rho(g^{-1})$, or a homomorphism from the opposite group.
4. Approve the character route for P07, with its characteristic-0 restriction and its status `conditional-on-exercises`.
5. Choose where the manifest lives: `arend-lib/docs/fulton-harris/` as in the whole-book plan, or the

repository's `docs/`.

Revision note. Corrections in this version

This version incorporates a review of the first draft.

- L08 is split into L08a and L08b. Image splitting is false over a general `SmithDomain`.
- D3 is now a pilot-only notion with no decidable equality; the general design is D3b.
- A constructive-$\mathbb{C}$ bridge for S_3 (R8a, R8b, L22) is added.
- The direction of the P01 homomorphism is corrected, and L02 is no longer called a monoid map.
- R3 now goes through the scalar commutant and Maschke averaging (L21).
- The exit criteria now have two tiers.
- Second review: L21 requires positive rank and reuses `LocalRing.connected`; `IsFinIrreducible` requires `NonZeroRing`, which excludes the zero ring; waves list their internal dependency chains; L19 moves to wave 2; R8 is split into a structural stage (R8a) and a character stage (R8b); and the claim about complex kernels and images is softened.
- Third review: $0 \neq 1$ is described as a nontriviality assumption; stale "any `Ring`" wording is corrected; R2 depends only on D3 and L18.
- Wave 1 implementation: the old notion is named `IsStronglyIrreducible`, with `SchurLemma-strong`. A nested name `IsIrreducible.Strong` would need a definition called `IsIrreducible`, and that name is reserved for D3b. The new Schur lemma concludes `IsEquiv f`, as the old one did.
- **Still deferred:** D3b, the general-module replacement for irreducibility. Completing this pilot validates finite-dimensional irreducibility only.

11 Implementation status (12 October 2026)

All P-, L- and R-items were implemented in `arend-lib`. The only open goals in the library outside block comments are the two registered exercise holes. Paths are relative to `arend-lib/src/Algebra/`.

Item	Declarations (module)	Notes
D3	`SplitSubLRepres`, `SplitSubLRepres.fromInvariant`, `IsFinIrreducible` (`Group/Representation/SplitSubLRepres`, `Irreducible`)	Old notion renamed `IsStronglyIrreducible`, Schur as `SchurLemma-strong`. The first component of `IsFinIrreducible` is stated as existence of a basis of positive length. `SplitSubLRepres` is an intertwiner that is a `SplitMono` in `LModuleCat`; it and its consumers are universe-polymorphic.
P01	`toUnitsHom`, `fromUnitsHom` and both round-trip laws (`Hom`); `PermRepr-not-FinIrreducible` (`PermutationRepresentation`); `rank-one-idempotent`, `rank-one-irreducible` (`Irreducible`)	Uses `InvGroup` from `Arith/Nat/EulerTotient`. Rank-one irreducibility holds over every `NonZeroCRing` without nontrivial idempotents (`IsConnected`), without averaging.
P01b	`IsStronglyIrreducible.excludedMiddle`	Proved; no axiom.
P02	`HomLRepres`, `DualLRepres`, `invariantHom`	`invariantHom` is the hole `FH_L01_Ex1_2`.
P03	`SchurLemma`	Over `DiscreteField`; concludes `IsEquiv f`.
P04	`splitProjector`, `maschke-split` (P04a, `Averaging`); `maschke-complement` (P04b, `Complement`)	P04a over `CRing`; `maschke-split` makes `incl` a `SplitMono` in `RepresentationCat`; P04b weakened during cleanup from `DiscreteField` to `SmithDomain`.
P05	`perm3-complete-reducibility` (`Symmetric3`)	Restricted to the presented S_3 input.
P06	`std-commutant`	Holds over every `CRing`: the equations need no division (the plan expected divisors of 6).
P07	`multiplicities` (`Multiplicities`)	Character route; characteristic 0 as a hypothesis; conditional on Ex. 1.2.
P08	`character`, `character-char`, `character_conj`, `character_ide` (`Characters`)	Basis independence via L04.
P09	`character_sum`, `character_hom`, `character_dual` (`CharacterFormulas`), `character_tensor` (`Tensor`)	Exterior square still deferred.
P10	`permCharacter`	Hole `FH_L02_Ex2_5`; no consumers.
P11	`invProj`, `invProj.trace`, `dim-invariants`, `invariants-split` (`Invariants`)	Dimension formula over `CRing`; existence over `DiscreteField`.
P12	`orthogonality-noniso`, `orthogonality-self`, `orthogonality-iso` (`Orthogonality`)	Conditional on Ex. 1.2.
L01–L05	`toLinearMap_*`, `toMatrix-action_*`, `Trace-coords`, `Trace-basis-indep`, `endoTrace` (`Module/TraceCoords`)	L04 holds over any `CRing`, for bases of possibly different lengths, with no inverses. L03 is subsumed by `Trace-coords`.
L06–L07, L11–L14	`CoordSystem`, `productCS`, `homCS`, `tensorCS`, `deltaCS`, `arrayCS` (`Module/CoordSystem`, `Tensor`)	One mechanism (finite set-indexed coordinates and `Trace-coords`) replaces block and Kronecker matrices.
L08–L09	`kernel-split`, `image-split`, `split-full-surj` (`Linear/SplitSubLModule`)	As corrected: kernel over `SmithDomain`, image over `DiscreteField`.
L10	`split-dim-<=` (`Linear/SplitSubLModule`)	No longer used downstream: rank-one irreducibility goes through `rank-one-idempotent`.

Item	Declarations (module)	Notes
L15	`trace-split`	Cyclicity of the trace; no kernel basis.
L16, L22	`natCoef-Inv-Rat`, `ratToComplex`, `natCoef-Inv-Complex` (`../Arith/Complex/RatHom`)	
L17–L20	`SymFinGroup`, `SymAction` (`Group/Symmetric/Action`), `FixedPointsFin`, `right-translate`	L20's inversion reindexing was not needed and was removed.
L21	`irreducible-from-commutant` (`Averaging`)	Uses `LocalRing.connected`.
R1–R7	`Symmetric3`, `Symmetric3Characters`	R1 is an explicit enumeration $\mathrm{Fin}\,6 \simeq S_3$; class sizes are not stated separately, since the sums run over all six elements. R6 covers all nine pairings, the reversed ones via the general lemma `pairing-symm`. Trivial and sign are irreducible over any `NonZeroCRing` without nontrivial idempotents (tested over $\mathbb{Z}$, also for $C_2 = S_2$).
R8a/b	`test/Algebra/Representation/S3Test.ard`	Irreducibility, decomposition, character values and pairings over `ComplexField`, with the only inverse being the image of 1/6.

Conditional declarations. Through `invariant-intertwining`, these depend on the hole `FH_L01_Ex1_2`:

- `orthogonality-noniso`, `orthogonality-self` and `orthogonality-iso`;
- `delta-case`, `count-pairing` and `multiplicities`.

Nothing consumes `FH_L02_Ex2_5`. The S_3 results R1–R8 do not depend on either hole.

Cleanup pass.

- 87 unused imports were removed. Each removal was kept only if the module still typechecked with the same goals.
- Dead helpers were deleted.
- Plan IDs in doc comments were replaced by book references.
- Hypothesis weakening was tried; P04b now holds over any `SmithDomain`.
- Not attempted: removing redundant explicit implicit arguments, and weakening `invariants-split` via a kernel splitting.

Completion audit.

- The manifest is `arend-lib/docs/fulton-harris/manifest.jsonl`: 45 records with formal targets, source relation and status. This settles decision 5.
- `arend-lib/tools/autoformalization/pilot_acceptance.py` checks in turn:
  1. every target exists;
  2. a fresh `-no-daemon -r -c` run, with targets in dependency order, reports no errors;
  3. the goals reported are exactly the registered holes;
  4. the test suite passes;
  5. the import-closure axiom audit is clean;
  6. the declaration-level consumer closure of the holes, computed with `-fu -json`, matches the manifest statuses;
  7. a negative control: the classical fixture is flagged;

8. a negative control: a temporary module with an invalid proof is rejected;
9. doc comments are well formed: multi-line docs use `{- | ... -}`, not a `-- |` line followed by `--` continuation lines, which would fall outside the documentation. Block docs must not contain `{-} or \code {-}`.

Check 1 queries the symbol index rather than the file text, so a definition swallowed by a stray comment delimiter is caught.

- A third control, inside the test suite, uses `fails`.
- Run on 12 October: `ACCEPTED`. A mutation of the manifest (marking P12 as `checked`) is rejected.
- Targets are checked in dependency order because, in a multi-module run, the CLI does not print the goals of a module first typechecked as a dependency.

Not yet done.

- The general-module design D3b.
- F2 below, needed before Lecture 3. F1 is implemented.

Follow-up items for Lectures 3 onward

Relation of the two irreducibility notions. An earlier version of `SplitSubLRepres` tested only free summands R^k with a linear retraction. The current notion tests every split subrepresentation (`SplitMono` in `LModuleCat`), and it is strictly stronger:

- **New implies old** over every `NonZeroRing`, via `fromInvariant`: a zero subrepresentation R^k has $k = 0$.
- **Old does not imply new** in general. Over $R = \mathbb{Z} \times \mathbb{Z}$, the trivial representation on R is irreducible in the old sense, but $R \times 0$ is a nonzero, non-surjective split subrepresentation. Hence `rank-one-irreducible` needs `IsConnected`.
- **They agree** over a `DiscreteField` (split the image of the idempotent $incl \circ retr$ by `image-split`). They also agree over a `LocalCRing` if finitely generated projective modules over local rings are free. That theorem is constructive in the literature (Lombardi–Quitté; exact statement to be checked), but it is not in arend-lib.
- Neither equivalence is formalized.

Over $\mathbb{C}$ the new notion is the more usable one. `ComplexField` is not a `DiscreteField`, so a basis of the image of an idempotent is not available. Isotypic projections, the projection formula (§2.4) and the decomposition of the regular representation all produce such images, and they are `SplitSubLRepres` values directly, with π as retraction.

F1 Basis-free character of a split subrepresentation

- **Problem.** `character` is defined for `FinLinRepres`, which needs a truncated basis. The summand `S.sub` of a `SplitSubLRepres` has none. Cor. 2.15 ($\langle \chi, \chi \rangle = 1$ implies irreducible), multiplicity counts and dimension counts need the characters of summands.
- **Proposal.** Define $\chi_S(g) = \mathrm{tr}_E(incl \circ g \circ retr)$ using `endoTrace` on E. Prove that it agrees with `character` when `S.sub` has a basis (trace cyclicity, as in L15). Prove additivity $\chi_E = \chi_S + \chi_C$ for the decomposition of `maschke-complement`.
- **Alternative.** Formalize "finitely generated projective modules over a `LocalCRing` are free". This would also close the gap between the two irreducibility notions, but it costs more.
- **Base.** Any `CRing`; no division needed.
- **Implemented** (`Group/Representation/SplitCharacter`, `Module/TraceCoords`):

- `endoTrace-comm` and `endoTrace-sum`: cyclicity and additivity of the basis-free trace;
- `splitCharacter`, defined with the record's retraction, and `splitCharacter-retr`: any other retraction gives the same value;
- `splitCharacter-basis`: if `S.sub` has a basis, the value is its ordinary `Character`. The proof expands `S.sub` along *retr* of a basis of E and applies `Trace-coords`, so no cyclicity across different modules is needed;
- `splitCharacter-sum`: $\chi_E = \chi_S + \chi_C$ for any splitting $e = \mathit{incl}_S(q\,e) + \mathit{incl}_C(q'\,e)$ with retractions q, q'; and `maschke-character`: over a `SmithDomain` with $|G|$ invertible, every split S has a complement C with $\chi_E = \chi_S + \chi_C$;
- test `std-splitCharacter-Q`: for the standard summand of the S_3 permutation representation over $\mathbb{Q}$, the split character equals `character (StdFin RatField)`.

Manifest record `GAP.F1`; acceptance run: `ACCEPTED`, 684 tests.

F2 Schur's lemma over $\mathbb{C}$

- **Problem.** `SchurLemma` is proved over `DiscreteField`, using `kernel-split` and `image-split`. Over `ComplexField`, the kernel and image of an intertwiner are not known to be split, so the lemma does not apply. The pilot avoided this over $\mathbb{C}$: it used the commutant route (L21) and explicit S_3 computations. From Lecture 3 on, Schur over $\mathbb{C}$ is used throughout.
- **This holds for both irreducibility notions.** It is not caused by the redesign.
- **Directions to evaluate.**
  - A conclusion weaker than "zero or isomorphism" that suffices for the character theory, for example a statement about the averaged projections.
  - Hypotheses that make kernels and images split, such as an apartness-based rank argument over a Heyting field.
  - Routing the needed consequences, such as orthogonality, through `maschke-split` and the commutant instead of through kernels.
- **Decision needed** before planning Lecture 3.